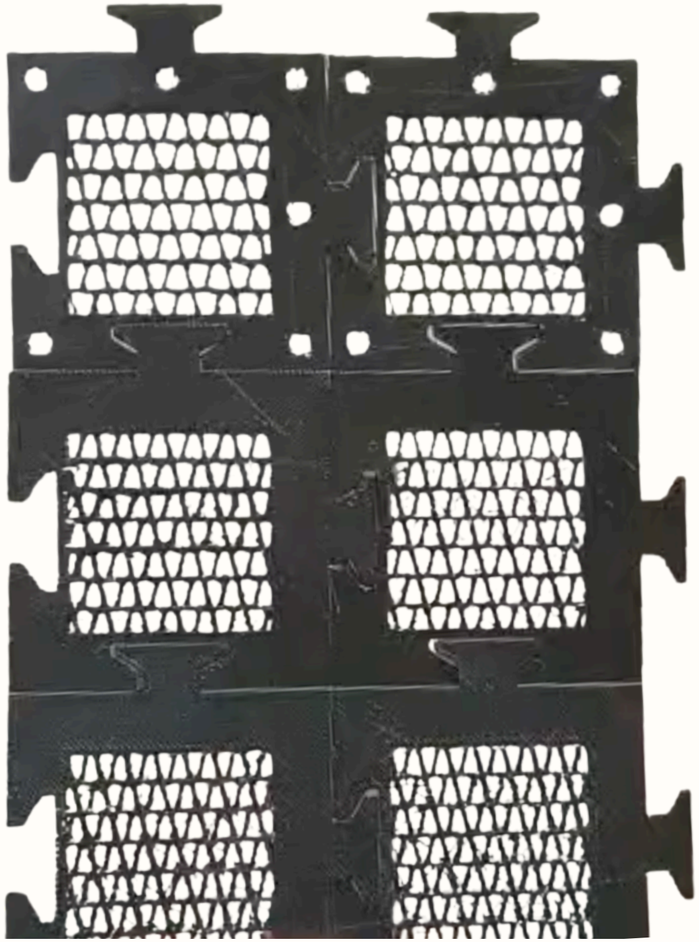
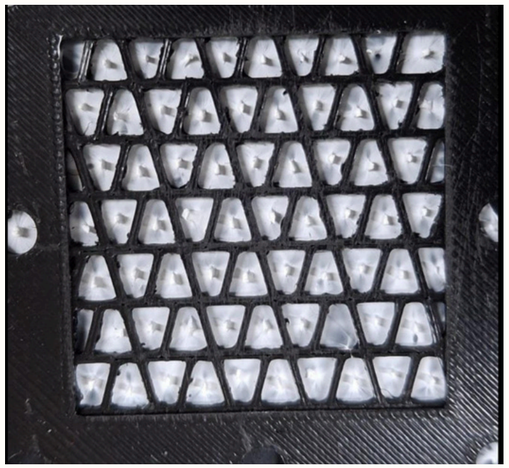
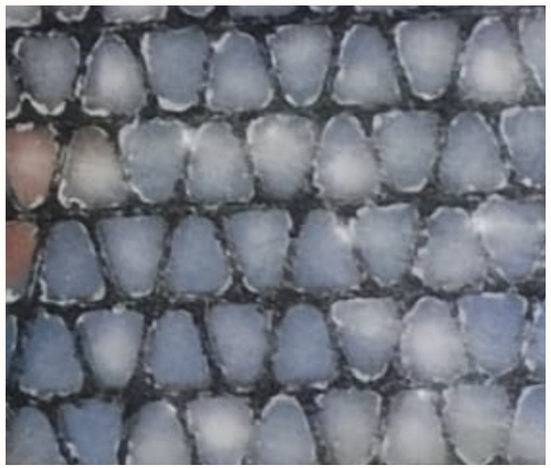
EcoFog: A Novel Composite Mesh for Passive Water Collection from Fog and High Relative-Humidity Environments







INTRODUCTION

2B+ People face water scarcity globally, especially in arid and semi-arid regions due to climate change, insufficient supply, and unaffordability.

UNICEF, “Water Scarcity,” UNICEF, 2020. https://www.unicef.org/wash/water-scarcity

25+ Countries have shown interest in fog harvesting, yet lack an efficient system.

Source: Pov, “Case Studies: Harvesting Water from the Sky | Thirst | POV | PBS,” POV | American Documentary Inc., Jan. 17, 2004. https://archive.pov.org/thirst/harvesting-water-from-the-sky/2/



Map showing global locations where fog harvesting has been successfully implemented & demonstrates potential for future development.

Source: K. Oktor, M. G. R. Dhuol, and M. E. Kalkan, “Fog harvesting: An effective solution to the water scarcity problem,” Sakarya University Journal of Science, vol. 28, no. 4, pp. 899–911, Aug. 2024, doi: 10.16984/saufenbilder.1480488.

A Few Places where Fog Harvesting has been tried:

India	Global
• Himachal Pradesh (IIT Mandi) • Maharashtra • Sikkim	• Morocco (Ait Baamrane) • Chile (Atacama Desert) • Peru

But, the poor efficiency hinders wide-scale implementation

LITERATURE REVIEW

1. Raschel Mesh (Deployed System)

- Field-implemented & low cost

Limitations

- Capture–drainage imbalance
- Non-biodegradable materials
- Low Water Collection efficiency



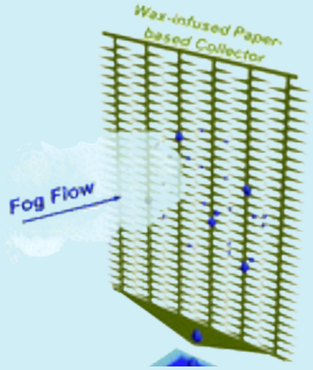
Source: M. Wei-Haas, “This device collects water from the clouds,” Smithsonian Magazine, Mar. 29, 2017. [Online]. Available: https://www.smithsonianmag.com/innovation/device-collects-water-clouds-180962705/

2. Biomimetics Mesh (Cactus Kirigami Structure)

- Bio-inspired architectures to enhance flux

Limitations

- Complex fabrication process
- Difficult to scale in under-resourced regions



Source: H. Bai et al., “Cactus kirigami for efficient fog harvesting: simplifying a 3D cactus into 2D paper art,” Journal of Materials Chemistry A, vol. 8, no. 27, pp. 13452–13458, Jan. 2020, doi: 10.1039/d0ta01204a.

3. Janus (Pore Engineered) Meshes

- Dual wettabilty allows simultaneous short-term drainage and capture

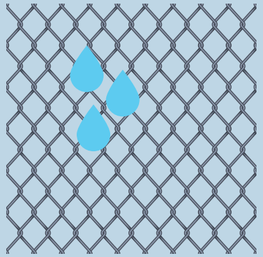
Limitations

- Prone to clogging
- Performs optimally only when fog flow aligns with its wettability gradient



Source: J. Song et al., “Sustainable Water Management with Photocatalytic Janus Mesh: Efficient Fog Harvesting, Water Purification, and Microbial Disinfection,” ACS Applied Materials & Interfaces, vol. 16, no. 44, pp. 61331–61343, Oct. 2024, doi: 10.1021/acsami.4c13466.

Unresolved Scientific Gaps



Low Capture

Source: B. S. Kennedy and J. B. Boreyko, “Bio-Inspired Fog Harvesting meshes: a review,” Advanced Functional Materials, vol. 34, no. 35, Nov. 2023, doi: 10.1002/adfm.202306162.



Complex Fabrication

Source: S. Nikbakht and M. M. Heyhat, “Experimental study of folded metal mesh for efficient fog harvesting,” Scientific Reports, vol. 15, no. 1, p. 19367, Jun. 2025, doi: 10.1038/s41598-025-04345-1.



Toxic, Non-Biodegradable Materials

Source: J. P. C. Gunarasan and J.-W. Lee, “Biomimetic materials for fog harvesting: Prospects and challenges,” Desalination, vol. 615, p. 119324, Aug. 2025, doi: 10.1016/j.desal.2025.119324.

OUR HYPOTHESIS

(1) Utilising electrospun Titanium Dioxide (TiO_2) nanoparticles (nanofillers) along with Polyvinyl alcohol (PVA) and Polyethylene oxide (PEO) nanofibers can capture high amounts of water droplets.

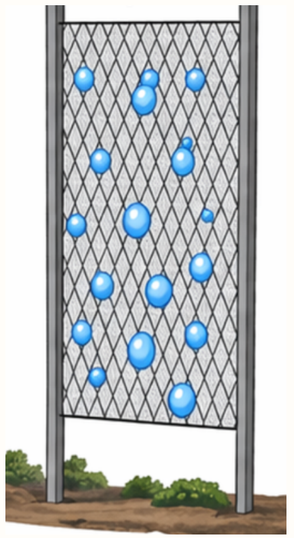
(2) Faster drainage of water droplets by utilising a hydrophobic Polylactic Acid (PLA) backbone may ultimately enhance the water collection rate through widely-used, non-toxic materials and scalable techniques.



Close-up of Mesh. *Photo taken by finalists, 2025.*

HIGH CAPTURE

Hydrophilic
Nanofibers for nucleation and coalescense

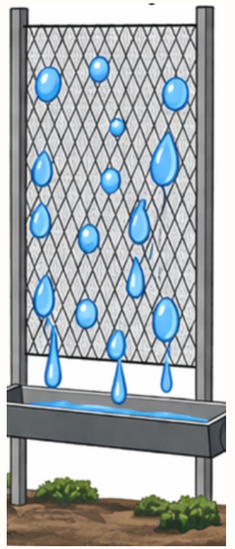




Close-up of Mesh. *Photo taken by finalists, 2025.*

FAST DRAINAGE

Hydrophobic
Backbone for droplet detachment and drainage

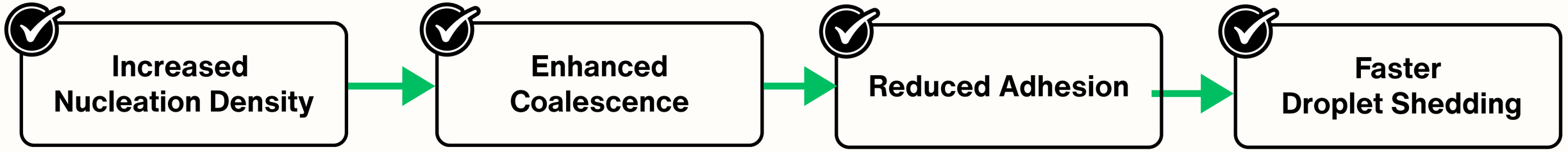


Hydrophilic matrix enhances droplet capture

TiO_2 protrusions provide additional nucleation sites

Hydrophobic PLA backbone causes water droplets to drain off quickly

INNOVATION



Materials & Purpose

1. **Polylactic Acid (PLA) Backbone**

- Hydrophobic backbone supports controlled drainage
- Biodegradable

2. **Electrospun Polyvinyl Alcohol (PVA) & Polyethylene Oxide (PEO) Nanofiber Matrix**

- PVA - Hydrophilic and lowers the energy barrier for fog droplet nucleation and capture
- PEO - Improves jet stability

3. **Titanium Dioxide (TiO_2) Nanoparticles (Nanofiller)**

- Forms hydrophilic protrusions on nanofibers that seed droplet growth

4. **Silicon Dioxide (SiO_2) Nanoparticles (Nanofiller)**

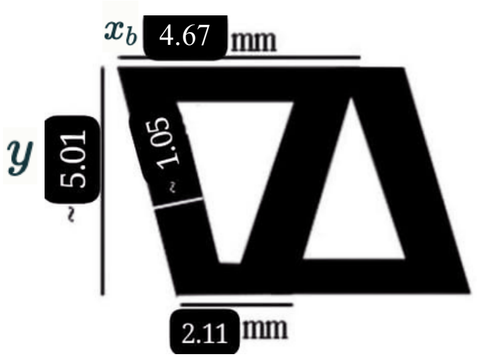
- Increases tensile strength of the brittle electrospun matrix

PLA RASCHEL-STYLE BACKBONE: DESIGN & 3D PRINTING

Shade Coefficient Calculation [11]

$$s = 1 - \frac{y(x_s + x_b)(b - t)}{2b(y + t)\left(t + \frac{x_s + x_b}{2}\right)}$$

3D printed Mesh Air Gap Dimensions:

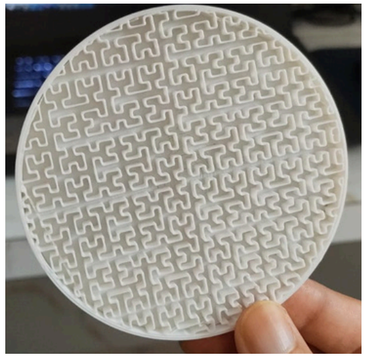


$$s = 1 - \frac{5.01 \times (2.11 + 4.67) \times (35 - 1.05)}{2 \times 35 \times (5.01 + 1.05) \times \left[1.05 + \frac{2.11 + 4.67}{2}\right]}$$

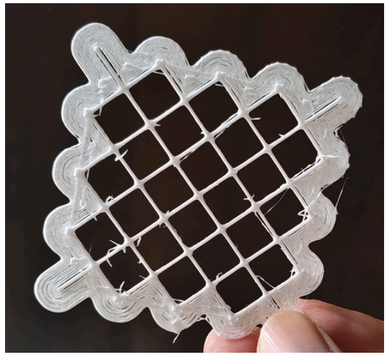
Shade coefficient = 0.388 (38.8%)

Fog harvesting studies report **33–50% as optimal**

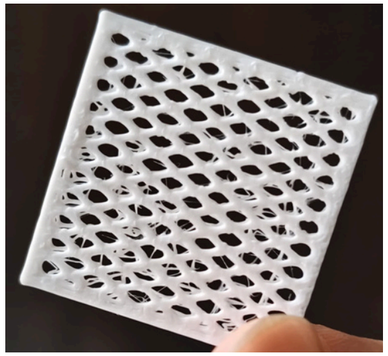
Early Prototypes and experiments with various air gap designs



Initial Backbone Design 1. *Photo taken by finalists, 2025.*



Initial Backbone Design 2. *Photo taken by finalists, 2025.*



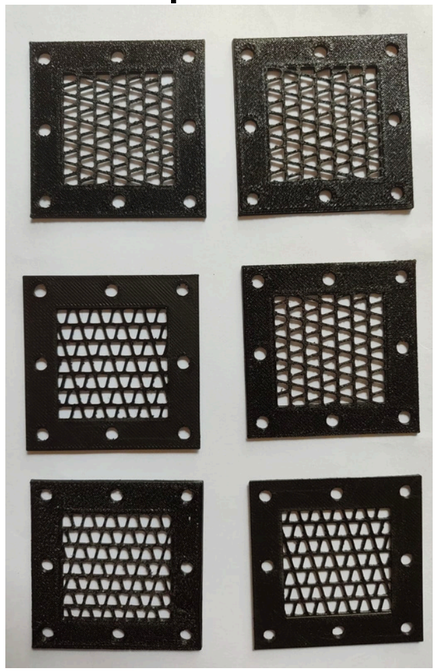
Initial Backbone Design 3. *Photo taken by finalists, 2025.*

CAD modeling of trapezium unit cells using Solidworks

3D printing using FDM (Fused Deposition Modeling)

- **Material**: Polylactic Acid
- **Nozzle temperature:** 200-210°C
- **Bed temperature:** 50-60°C
- **Layer height:** 0.2 mm (precision resolution)
- **Print time**: ~30 minutes per mesh
- **Post-processing**: Minimal support removal, no post-curing needed

Final Optimized Mesh



Final Backbone Design. *Photo taken by finalists, 2025.*

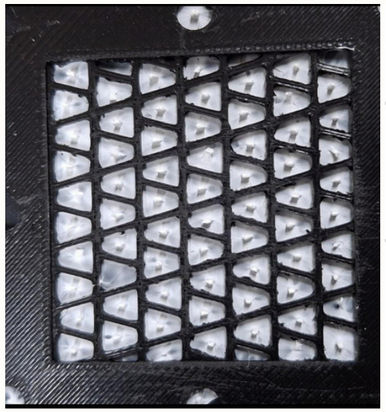
ELECTROSPINNING

Electrospinning Sample Preparation (PVA, PEO, TiO_2, SiO_2)



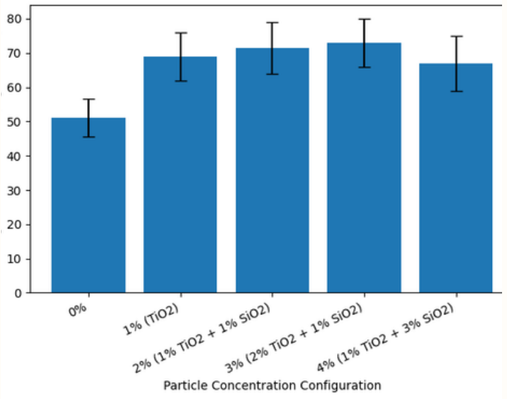
Electrospinning process for sample preparation. *Photo taken by finalists, 2025.*

Electrospun matrix and nanofillers on mesh



Electrospun Matrix on Backbone. *Photo taken by finalists, 2025.*

Nanoparticle Concentration Optimization



Effect of Particle Concentration on Water Collection Rate. *Graph created by finalists using Matplotlib, 2026.*

CROSSLINKING

- Improves resistance to water
- Enhances mechanical strength and durability to withstand environmental stress

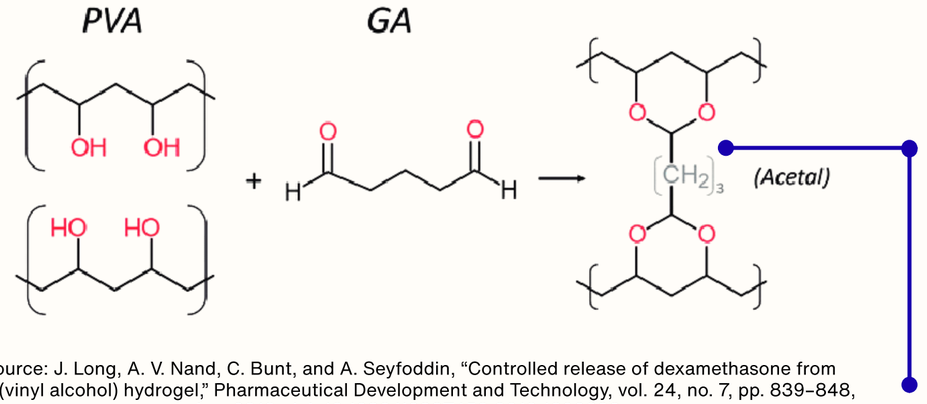


Crosslinking of the Electrospun Mesh by Glutaraldehyde Vapours. Photo taken by Finalists, 2025.

Needle-to-collector distance optimization



Deposition of Nanofibers at Different Distances. *Photo taken by Finalists, 2025.*



Source: J. Long, A. V. Nand, C. Bunt, and A. Seyfoddin, “Controlled release of dexamethasone from poly(vinyl alcohol) hydrogel,” Pharmaceutical Development and Technology, vol. 24, no. 7, pp. 839–848, Apr. 2019, doi: 10.1080/10837450.2019.1602632.

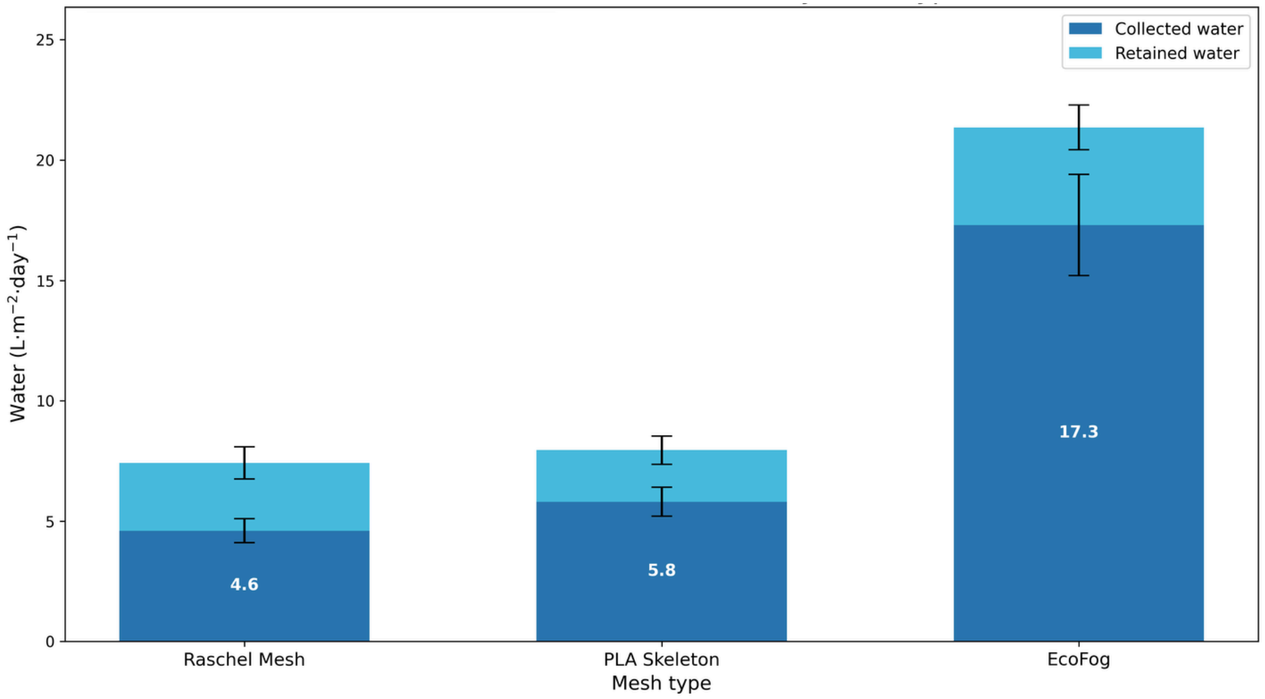
Crosslinking forms covalent acetal bonds between PVA chains as glutaraldehyde reacts with hydroxyl groups

WATER COLLECTION RATE

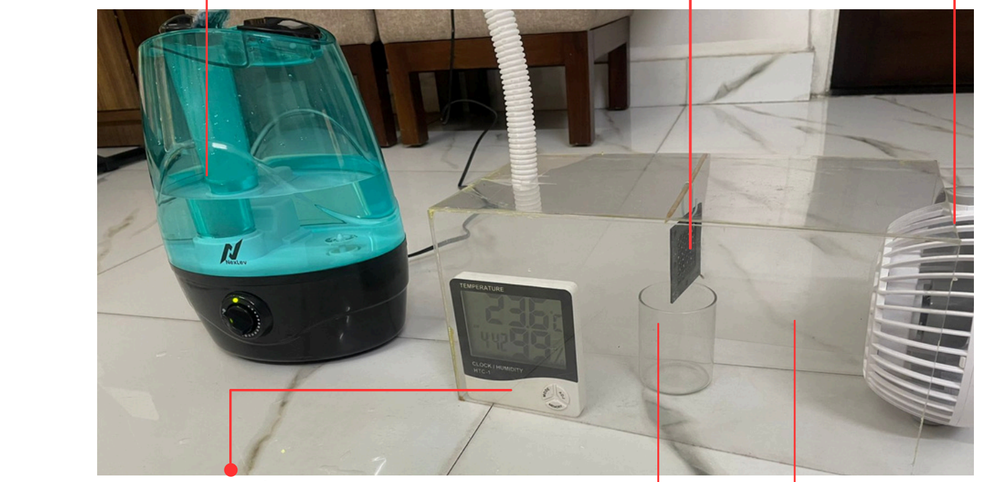
Calculation of Dependent Variable

$$\text{Water collection rate} = \frac{\text{Mass of water collected}}{\text{Effective mesh area} \times \text{Time}}$$

Water Collection (L m^{-2} day^{-1}) & Retained-Water Fraction Comparison



Water Collection (L m^{-2} day^{-1}) and Retained Water on Mesh for EcoFog, Pure PLA backbone, and Raschel Mesh. *Graph created by finalists using Matplotlib, 2026.*



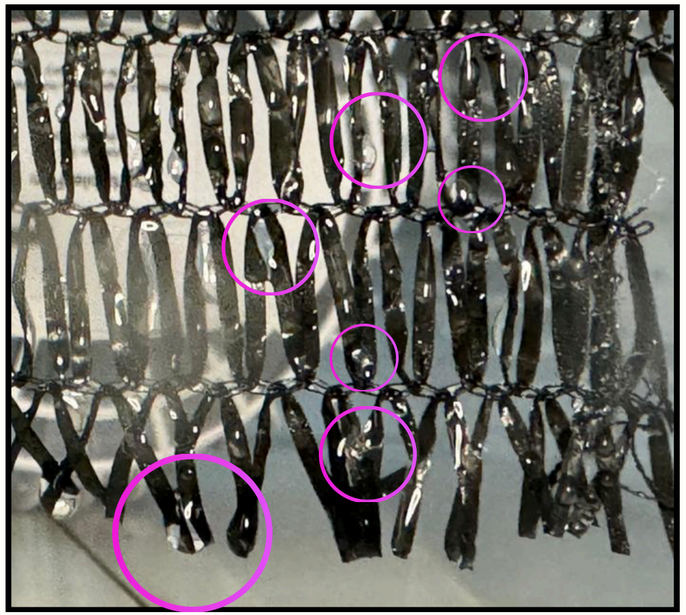
Fog Simulation Chamber Components. *Photo taken by finalists, 2025.*

Control Variables

- **Fog Generation Rate**
- **Exposure Time (4 hours)**
- **Airflow / Wind Speed (3.0 m/s)**
- **Chamber Conditions (99±1% RH, 20±2°C)**
- **Mesh Orientation and Position**

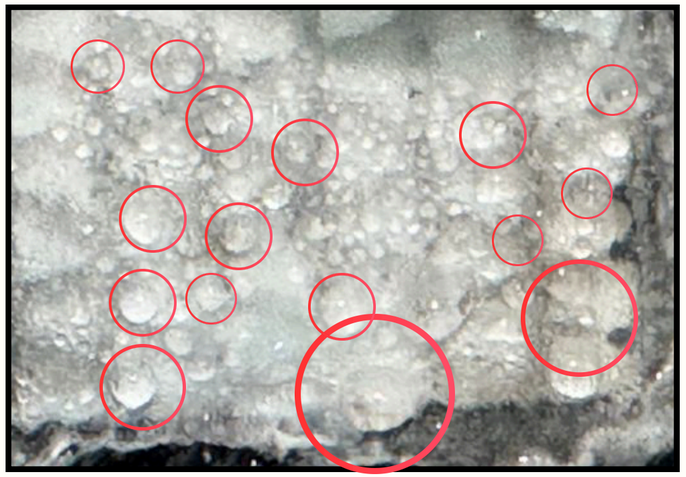
DROPLET FORMATION

Raschel Mesh



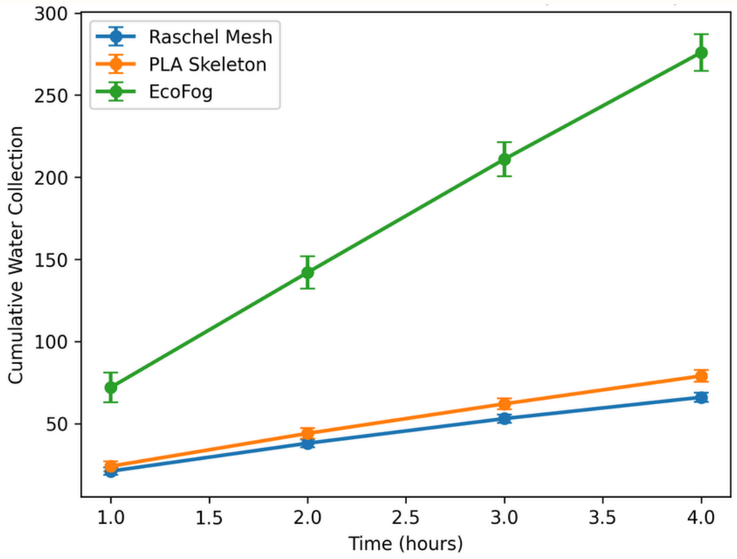
Droplet Formation for Rashcel Mesh. *Photo taken by Finalists, 2025.*

EcoFog



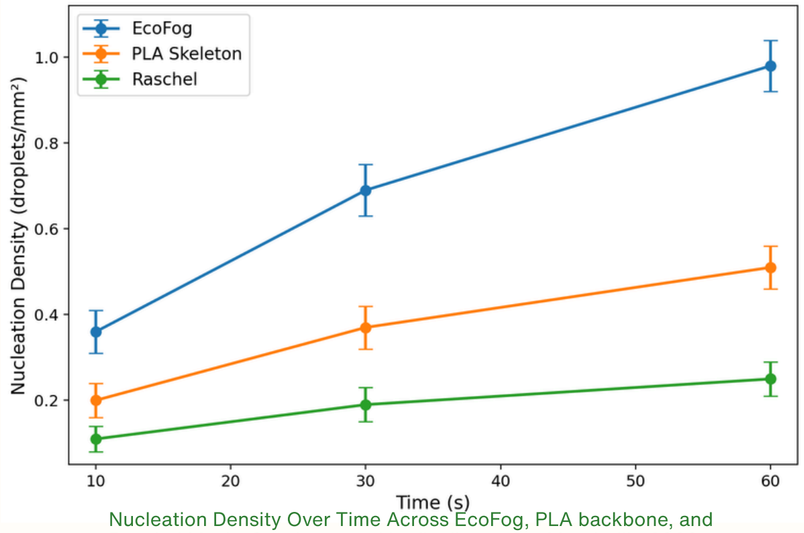
Droplet Formation for EcoFog. *Photo taken by Finalists, 2026.*

DATA REPRESENTATION



Cumulative Water Collection graph (mg cm^2) Across EcoFog, PLA backbone, and Raschel Mesh. *Graph created by finalists using Matplotlib, 2026.*

Cumulative Water Collection for 4 hours Comparison (mg/cm^2)

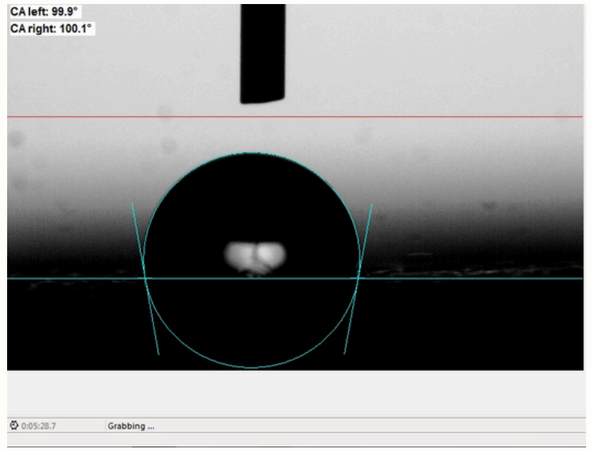


Nucleation Density Over Time Across EcoFog, PLA backbone, and Raschel Mesh. *Graph created by finalists using Matplotlib, 2026.*

Nucleation Density Comparison (droplets/mm^2)

CONTACT ANGLE TESTING (SESSILE DROP)

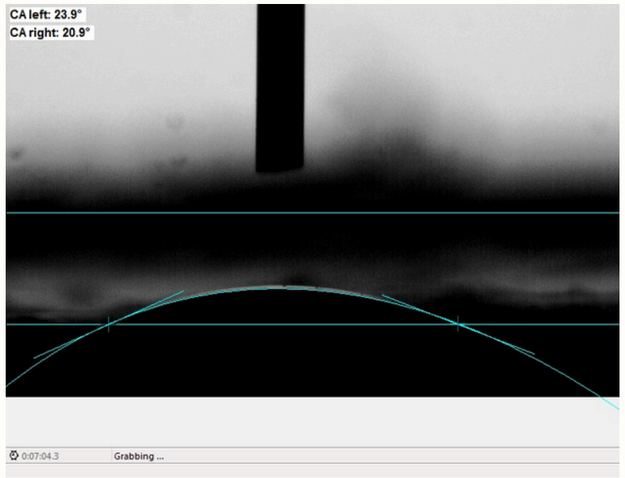
PLA Backbone



High static contact angle (~100°), indicating **hydrophobicity**

Contact Angle of PLA Backbone. *Image taken by finalists, 2026.*

Nanofiber + Nanoparticle Matrix



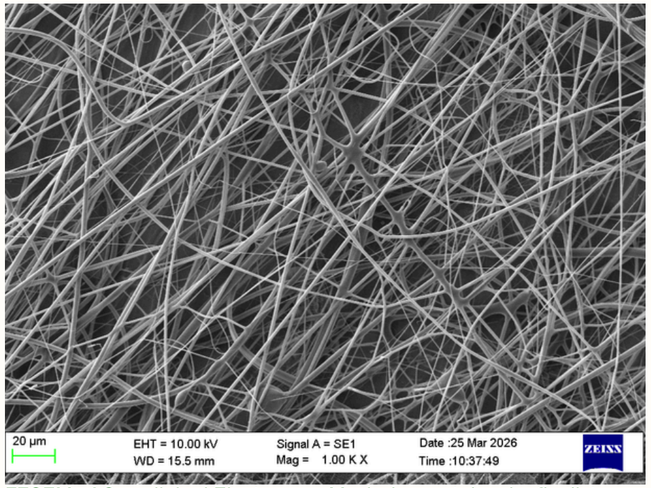
Very low contact angle (20.9°–23.9°), indicating **hydrophilicity**

Contact Angle of Electrospun Mesh. *Image taken by finalists, 2026.*

FESEM CHARACTERIZATION

Nanofiber + Nanoparticle Matrix

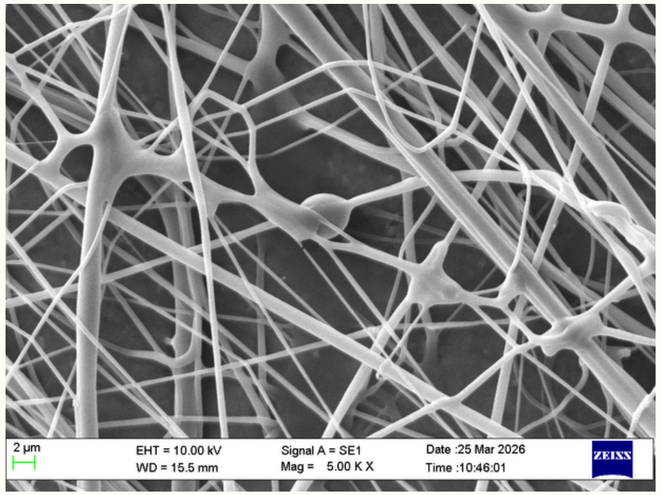
Uniform and **bead-free** nanofibers at optimal concentrations (1 wt% of TiO_2 and 1 wt% SiO_2 with base matrix)



FESEM of Crosslinked Electrospun Mesh. *Image taken by finalists, 2026.*

2 wt% TiO_2 and 1 wt% SiO_2 with base matrix

Agglomeration and frequent **bead formation** observed in the non-uniform nanofibers at higher concentrations of polymers and nanofillers



FESEM of Crosslinked Electrospun Mesh (2 wt% TiO_2 & 1 wt% SiO_2. *Image taken by finalists, 2026.*

FIELD DEPLOYMENT AND APPLICATION

40 m^2 Mesh Water Daily Production

~600

Litres

- **3.46x** higher than traditional Raschel mesh
- Sufficient to meet WHO minimum daily water requirements (50 $L \cdot person^{-1} \cdot day^{-1}$) for 12 individuals

Source: W. H. Organization, Guidelines for drinking-water quality: fourth edition incorporating the first and second addenda. World Health Organization, 2022.

Scope of Global Deployment

- **Coast Redwood Belt:**
  - Redwood regions, running from southern Oregon down to central California
  - Summertime fog
- **Indo-Gangetic Plains:**
  - High volume of fog especially during winters
- **Hilly regions**
- **Coastal zones**

Source 1: J. Petreshen, S. F. Dymond, E. T. Keppeler, S. T. Allen, and J. W. Wagenbrenner, “Fog presence and ecosystem responses in a managed coast redwood forest,” Agricultural and Forest Meteorology, vol. 368, p. 110525, Jun. 2025, doi: https://doi.org/10.1016/j.agrformet.2025.110525.

Source 2: [G. River, “Redwoods Rising to the Sky - Friends of Gualala River,” Friends of Gualala River, Nov. 11, 2021. https://gualalariver.org/forestry/redwoods-rising-to-the-sky/

Source 3: G. K. SAWAISARJE, P. Khare, C. Y. SHIRKE, S. DEEPAKUMAR, and N. M. NARKHEDE, “Study of winter fog over Indian subcontinent : Climatological perspectives,” MAUSAM, vol. 65, no. 1, pp. 19–28, Dec. 2021, doi: https://doi.org/10.54302/mausam.v65i1.858.

KEY REFERENCES

[1] UNICEF, “Water scarcity.” https://www.unicef.org/wash/water-scarcity
[2] K. Oktor, M. G. R. Dhuol, and M. E. Kalkan, “Fog harvesting: An effective solution to the water scarcity problem,” Sakarya Univ. J. Sci., vol. 28, no. 4, pp. 899–911, 2024. https://doi.org/10.16984/saufenbilder.1480488
[3] R. Emerine and S. Chou, “Fast delivery of melatonin from electrospun blend polyvinyl alcohol and polyethylene oxide (PVA/PEO) fibers,” AIMS Bioeng., vol. 9, no. 2, pp. 178–196, 2022. https://doi.org/10.3934/bioeng.2022013
[4] M. Balbaşı and B. Gözütok, “Poly(vinyl alcohol)-colloidal silica composite membranes for fuel cells,” Synth. Met., vol. 160, no. 1–2, pp. 150–155, 2010. https://doi.org/10.1016/j.synthmet.2009.10.023
[5] W. K. Ng, C. Y. Wong, N. A. H. Rosli, K. V. Kannan, K. S. Loh, B. L. Chua, and W. Y. Wong, “The impact of double crosslinking and alkaline activation strategies on the multifaceted characteristics of quaternized poly(vinyl alcohol) anion exchange membranes,” Energy Adv., vol. 4, no. 3, pp. 400–413, 2025. https://doi.org/10.1039/d4ya00555d
[6] United Nations, “Water.” https://www.un.org/en/global-issues/water
[7] WWF, “The lifecycle of plastics.” https://wwf.org.au/blogs/the-lifecycle-of-plastics/
[8] D. Rosa et al., “Titanium dioxide nanoparticles doped with iron for water treatment via photocatalysis: A review,” Nanomaterials, vol. 14, no. 3, p. 293, 2024. https://doi.org/10.3390/nano14030293
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[10] R. P. Shaikh et al., “Crosslinked electrospun polyvinyl alcohol nanofibrous membranes: Elucidation of their physicochemical, physicomechanical, and molecular disposition,” Biofabrication, vol. 4, no. 2, Art. no. 025002, 2012. https://doi.org/10.1088/1758-5082/4/2/025002
[11] A. A. Elshennawy et al., “Evaluating mesh geometry and shade coefficient for fog harvesting collectors,” Water Resour. Manage., 2023. https://doi.org/10.1007/s11269-023-03644-
[12] A. Katselas et al., “Quantification of nucleation site density as a function of surface wettability on smooth surfaces,” Adv. Mater. Interfaces, vol. 9, no. 15, Art. no. 2200246, 2022. https://doi.org/10.1002/admi.202200246